



SilkSol AI

Parametric Cargo Delay-Cover Protocol for
the Trans-Caspian Middle Corridor (TITR)

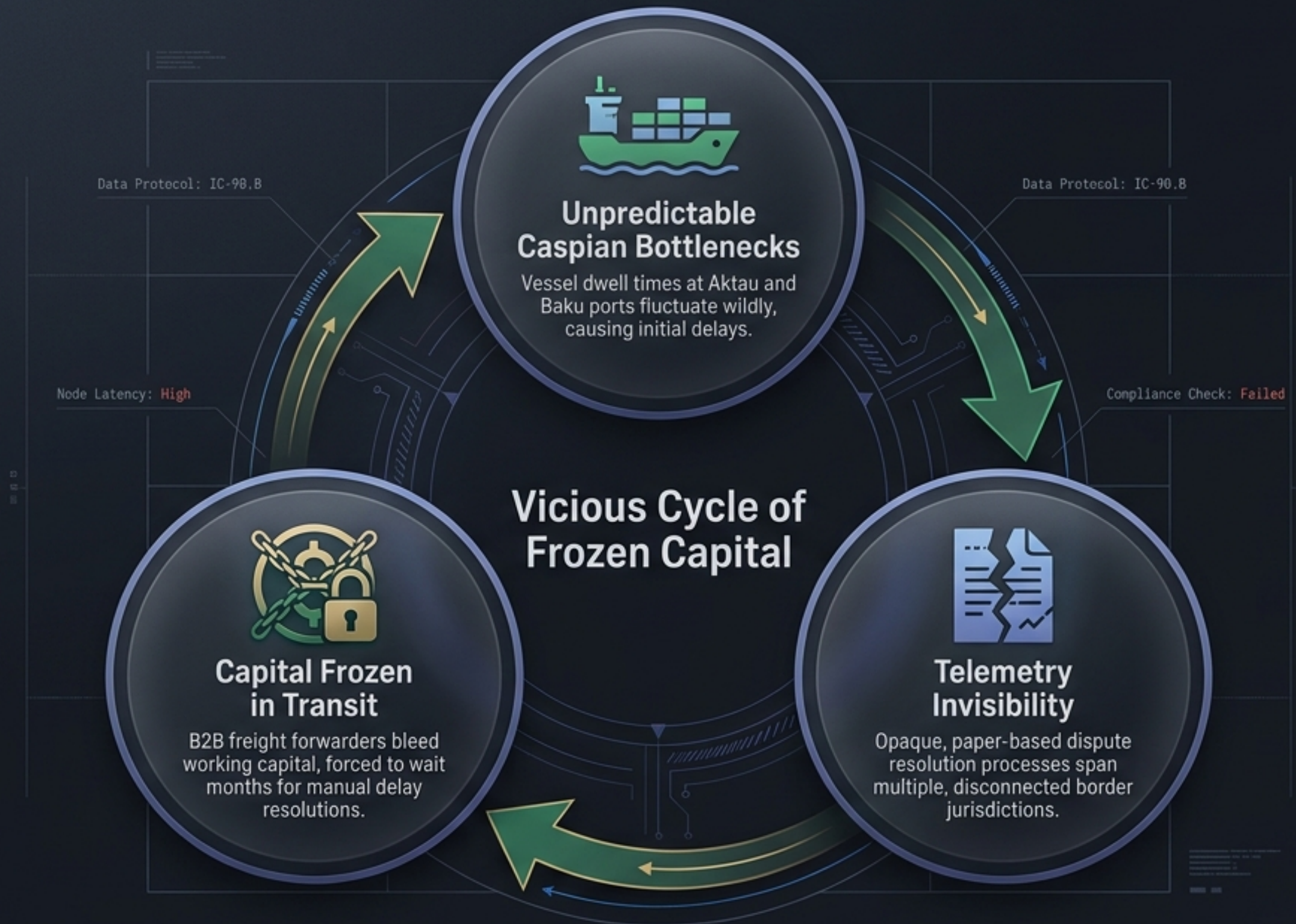
Built on
Solana Devnet

Colosseum Frontier
Hackathon Entry

AIFC Regulatory
Sandbox Concept

Founder: Sophia Akhmetova – Solo Founder & Engineer (Aktau, Kazakhstan)

Middle Corridor Logistics is Crippled by 90-Day Claims Bureaucracy

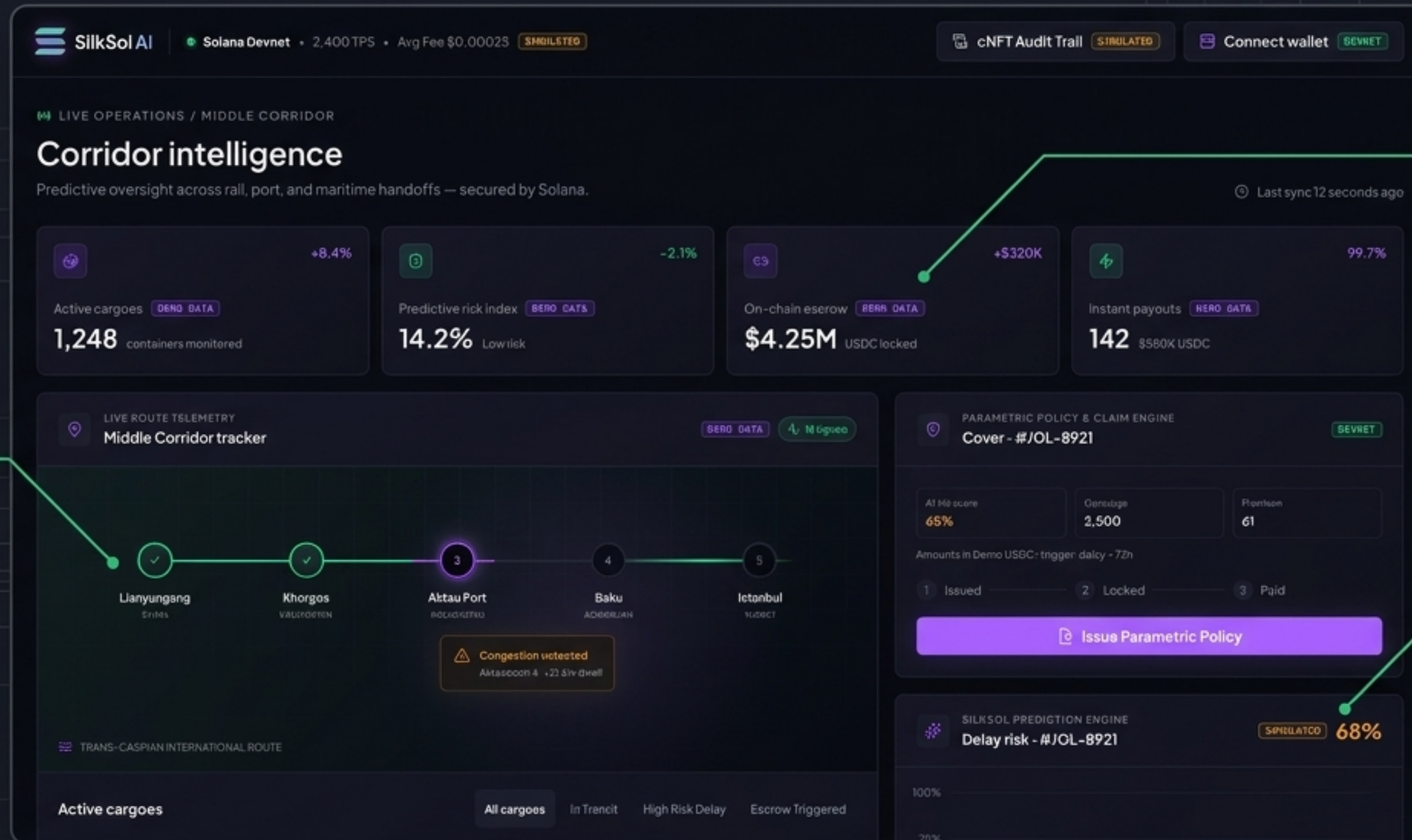


Automated Parametric Risk Mitigation for Caspian Freight



Rule-based, sub-second escrow release entirely bypassing paper bureaucracy.

Corridor Intelligence Secured by Solana

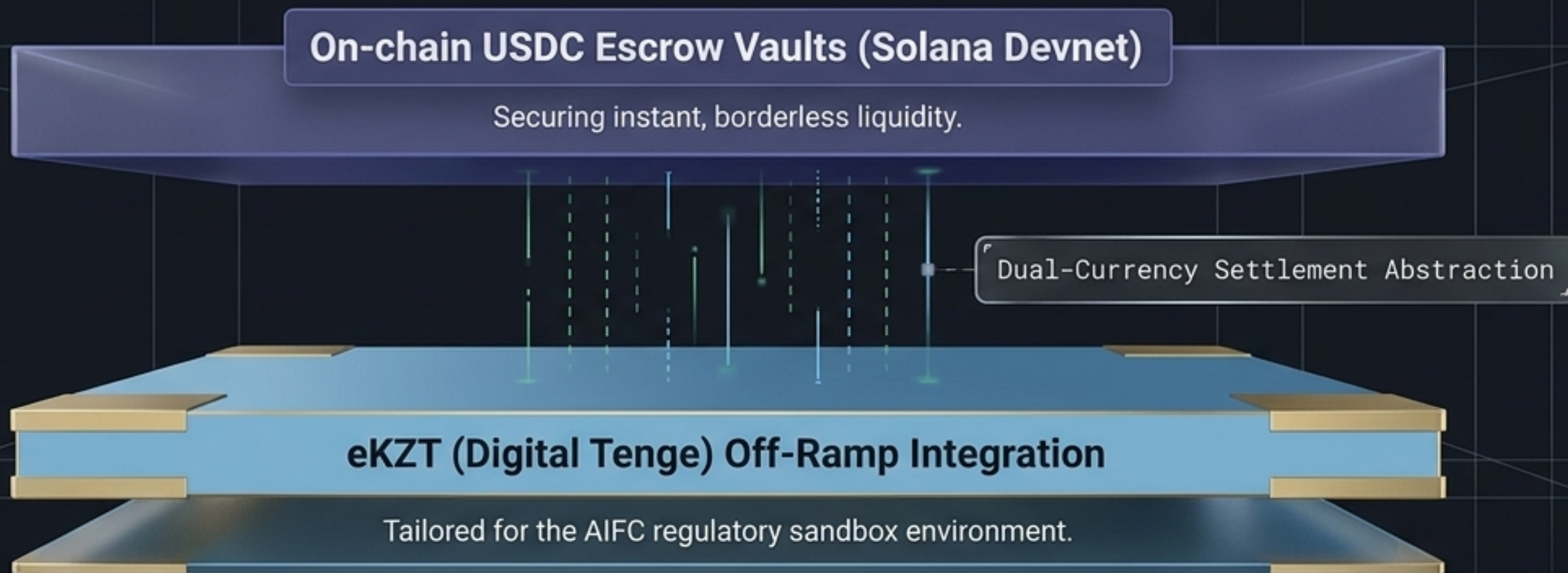


On-Chain
Escrow

Live Route
Telemetry

Predictive
Risk Index

Bridging Solana Liquidity with Kazakhstani B2B Compliance



Zero paperwork



Full tax alignment



End-to-end legal auditability for Kazakhstani logistics enterprises.

Built for High-Volume Supply Chain Telemetry

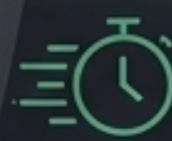
COST



\$0.00002
micro-fees

Economical enough to enable continuous, high-frequency cargo event logging without network cost decay. (Replaces \$10+ L1 gas fees).

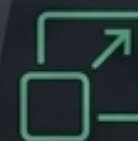
SPEED



~400ms
Finality

Sub-second block finality ensures instant smart contract state execution when parametric thresholds are triggered.

SCALE



State
Compression

cNFT integration allows the protocol to log millions of sequential cargo status updates on-chain economically.

Real Domain Context from the Port of Aktau



The Heartland of Trans-Caspian Trade

Developed directly on the ground in Aktau. SilkSol embeds direct domain insights into port operations, railway logistics, and regional B2B friction points that remote engineering teams cannot access.

Collapsing Claims Settlement Times Programmatically

Traditional Insurance

60-90 days manual adjustment.

Heavy legal overhead and paperwork.

Subjective dispute resolution.

SilkSol Protocol

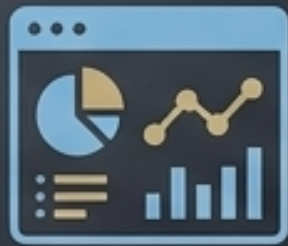
Sub-second programmatic escrow release upon verified oracle trigger.

0% paper bureaucracy.

100% verifiable cNFT audit trail.

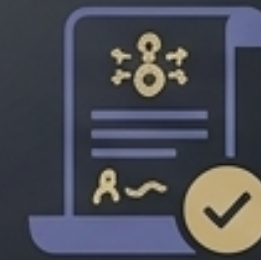
Sustainable Hybrid B2B Revenue Architecture

Enterprise SaaS Subscription



Recurring operational fee for real-time predictive risk analytics and telemetry dashboard access.

Protocol Settlement Fee



0.5% protocol fee captured directly from automated, on-chain escrow releases.

Value
Capture

From Caspian Sandbox to Eurasian Trade Corridors

Targeting the core Trans-Caspian Route (TITR) pilot (Aktau-Baku corridor).

Executing within the secure AIFC Sandbox testing environment.

Phase 1: Active MVP & Sandbox Pilot

Phase 2: Global Network & Integration

Expansion to wider Eurasian multimodal routes.

Deep integration with enterprise ERP systems for automated cross-border trade.

Deep Domain Context & ML Risk Analytics



**Sophia
Akhmetova**

[Applied ML & Predictive Risk Analytics Engineer]

[Oilfield Services Background]

[Kazakhstan Industrial Domain Expert]

[Solo Solana Builder (Aktau, KZ)]

Combines hands-on Kazakhstani industrial domain knowledge with specialized predictive AI modeling and practical Web3 architecture.

Catalyzing the Caspian Pilot

- ☒ Technical Mentorship
- ☒ Ecosystem Partnerships
- ☒ Grant Support (Solana Colosseum & Superteam Kazakhstan)